3/4.7.10 CHILLED WATER SYSTEM - AUXILIARY BUILDING SUBSYSTEM

LIMITING CONDITION FOR OPERATION

3.7.10 The chilled water system loop which services the safety-related loads in the Auxiliary Building shall be OPERABLE with one of the following configurations:

	a	b	c
Configuration	1. Three OPERABLE chillers and, 2. Two OPERABLE chilled water pumps	1. Two OPERABLE chillers and, 2. Two OPERABLE chilled water pumps	1. Three OPERABLE chillers and, 2. Two OPERABLE chilled water pumps from either Unit 1 or Unit 2 (Units Cross-tied)(2)
APPLICABILITY	1. ALL MODES and during movement of irradiated fuel assemblies	1. From November 1 through April 30 in ALL MODES and during movement of irradiated fuel assemblies# 2. The Unit 2 Emergency Control Air Compressor (ECAC) is isolated from the chilled water system 3. Chilled water flow to the third chiller that is not in service is isolated(1) 4. Control Room Emergency Air Conditioning System (CREACS) alignment a. BOTH CREACS trains OPERABLE, no additional chilled water heat load removal required, OR b. Single CREACS train OPERABLE (TS 3.7.6 ACTION a.) the following restrictions apply: i. Alignment only permitted to Unit 1 ii. Unit 1 must be in the LCO 3.7.10a configuration iii. Non-essential heat loads are isolated from the chilled water system on BOTH Units	1. From November 1 through April 30 in ALL MODES and during movement of irradiated fuel assemblies## 2. The Unit 1 and Unit 2 ECACs are isolated from the chilled water system 3. Non-Essential heat loads are isolated from the chilled water system on BOTH Units 4. BOTH CREACS trains are operable per TS 3.7.6 (single filtration train alignment is not permitted) 5. Unit chilled water cross-tie valves are OPEN 6. Administrative controls are in place for the Unit providing the required components to notify the other Unit if a chiller or pump becomes inoperable

\# The LCO 3.7.10b configuration may only be used for periods of 60 contiguous days. The 60-contiguous days does not apply for LCO 3.7.10b entry to support the replacement of all 6 original chillers (Units 1 and 2).

\## The LCO 3.7.10c configuration may only be used for periods of 45 contiguous days.

LIMITING CONDITION FOR OPERATION (Continued)

ACTION[3]: MODES 1, 2, 3, and 4

a. With one of the required chillers inoperable:

1. Remove[4] the appropriate non-essential heat loads from the chilled water system within 4 hours and;

2. Restore the chiller to OPERABLE status within 14 days or;

3. Be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.

b. With two of the required chillers inoperable[5][6]:

1. Remove the appropriate non-essential heat loads from the chilled water system within 4 hours and;

2. Align the control room emergency air conditioning system (CREACs) for single filtration operation using the Salem Unit 1 train within 4 hours and;

3. Restore at least one chiller to OPERABLE status within 72 hours or;

4. Be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.

c. With one chilled water pump inoperable, restore the chilled water pump to OPERABLE status within 7 days or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours

ACTION[3]: MODES 5 and 6 or during movement of irradiated fuel assemblies.*

a. With one of the required chillers inoperable:

1. Remove[4] the appropriate non-essential heat loads from the Chilled Water System within 4 hours and;

2. Restore the chiller to OPERABLE status within 14 days or;

3. Suspend CORE ALTERATIONS and movement of irradiated fuel assemblies.

LIMITING CONDITION FOR OPERATION (Continued)

b. With two of the required chillers inoperable(5)(6):

1. Remove the appropriate non-essential heat loads from the chilled water system within 4 hours and;

2. Align the control room emergency air conditioning system (CREACs) for single filtration operation using the Salem Unit 1 train within 4 hours and;

3. Restore at least one chiller to OPERABLE status within 72 hours or;

4. Suspend CORE ALTERATIONS and movement of irradiated fuel assemblies.

c. With one chilled water pump inoperable, restore the chilled water pump to OPERABLE status within 7 days or suspend CORE ALTERATIONS and movement of irradiated fuel assemblies.

SURVEILLANCE REQUIREMENTS

4.7.10 The chilled water loop which services the safety-related loads in the Auxiliary Building shall be demonstrated OPERABLE:

a. In accordance with the Surveillance Frequency Control Program by verifying that each manual valve in the chilled water system flow path servicing safety related components that is not locked, sealed, or otherwise secured in position, is in its correct position.

b. In accordance with the Surveillance Frequency Control Program, by verifying that each automatic valve actuates to its correct position on a Safeguards Initiation signal.

c. In accordance with the Surveillance Frequency Control Program by verifying that each chillers starts and runs.

d. When in the LCO 3.7.10b configuration verify once per 24 hours:

(i) The Unit 2 ECAC is isolated from the chilled water system,
(ii) Chilled water flow is isolated to the third chiller that is not in service and,
(iii) If CREACS is in single filtration alignment verify non-essential heat loads are isolated from the chilled water system on BOTH Units.

e. When in the LCO 3.7.10c configuration verify once per 24 hours:

(i) The Unit 1 and Unit 2 ECACs are isolated from the chilled water system,
(ii) Non-essential heat loads are isolated from the chilled water system and,
(iii) Cross-tie valves are verified OPEN.

Surveillance Requirements (Continued)

NOTES

(1) When transitioning from the LCO 3.7.10b to the LCO 3.7.10a configuration, the chiller may be un-isolated (restored to service) under administrative controls

(2) The LCO 3.7.10c (Cross-Tied) configuration is common to both Units; either Unit 1 chilled water components are required operable, OR Unit 2. A combination of both Units chilled water components is not permitted. When transitioning from the LCO 3.7.10c configuration to either the LCO 3.7.10a or LCO 3.7.10b configurations, chilled water components may be restored to service under administrative controls

(3) When in the LCO 3.7.10c configuration ACTIONs are applicable for both Units

(4) When in the LCO 3.7.10c configuration this ACTION has already been implemented

(5) When in the LCO 3.7.10b configuration, implement Action b.2 AND Action b.4 OR transition to the LCO 3.7.10c configuration

(6) When in LCO 3.7.10c configuration, proceed directly to Action b.4

* During Modes 5 and 6 and during movement of irradiated fuel assemblies, chilled water components are not considered to be inoperable solely on the basis that the backup emergency power source, diesel generator, is inoperable. This is not applicable to the LCO 3.7.10c configuration.